**2.Develop collaborative workflows using remote repository operations including cloning, fetching, rebasing, and custom merge commits to enable team-based development.**

- **Clone remote repositories to local machine**
- **Fetch latest changes and rebase local branches**
- **Merge feature branches with custom commit messages**

## Step 1: Clone the remote repository

```
BMSIT@DESKTOP-9VOIESP MINGW64 ~ (main)
$ git init
Reinitialized existing Git repository in C:/Users/BMSIT/.git/

BMSIT@DESKTOP-9VOIESP MINGW64 ~ (main)
$ git clone https://github.com/karbekk/firstGitLab.git
Cloning into 'firstGitLab'...
warning: You appear to have cloned an empty repository.

BMSIT@DESKTOP-9VOIESP MINGW64 ~ (main)
$ git clone https://github.com/karbekk/Machine-Learning-programs.git
Cloning into 'Machine-Learning-programs'...
remote: Enumerating objects: 87, done.
remote: Counting objects: 100% (27/27), done.
remote: Compressing objects: 100% (16/16), done.
remote: Total 87 (delta 19), reused 11 (delta 11), pack-reused 60 (from 1)
Receiving objects: 100% (87/87), 56.19 KiB | 1.87 MiB/s, done.
Resolving deltas: 100% (25/25), done.
```

## Step 2: Create a feature branch and commit locally

```
BMSIT@DESKTOP-9VOIESP MINGW64 ~ (main)
$ git checkout -b feature-1
Switched to a new branch 'feature-1'

BMSIT@DESKTOP-9VOIESP MINGW64 ~ (feature-1)
$ echo "Hello from feature-1" > file1.txt

BMSIT@DESKTOP-9VOIESP MINGW64 ~ (feature-1)
$ git add file1.txt
warning: in the working copy of 'file1.txt', LF will be replaced by CRLF the next time Git touches it

BMSIT@DESKTOP-9VOIESP MINGW64 ~ (feature-1)
$ git commit -m "add file1"
[feature-1 ae078a8] add file1
 1 file changed, 1 insertion(+)
 create mode 100644 file1.txt
```

## Step 3: Push the feature branch

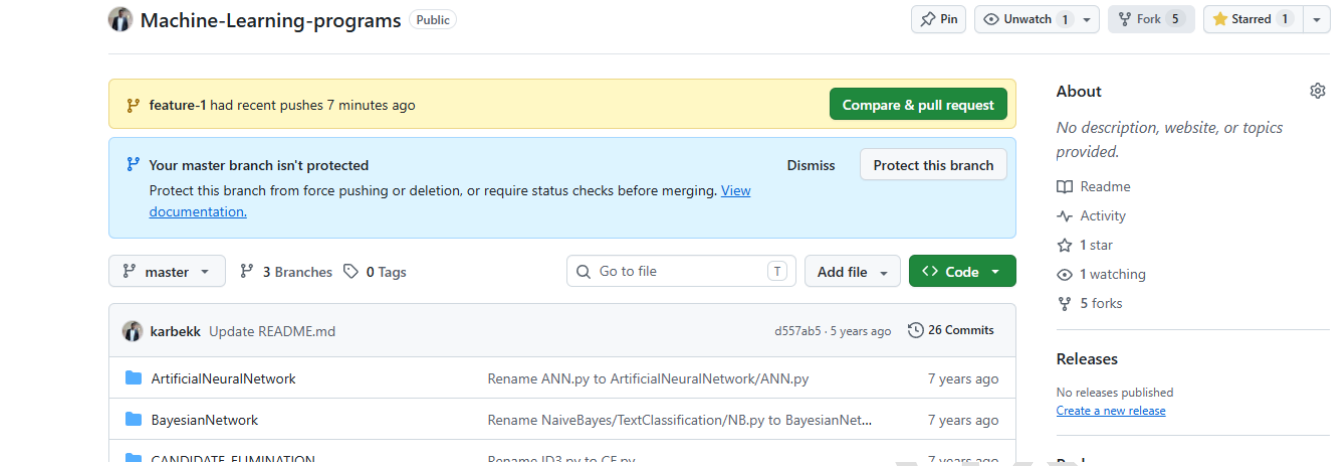
```
BMSIT@DESKTOP-9VOIESP MINGW64 ~ (feature-1)
$ git remote -v

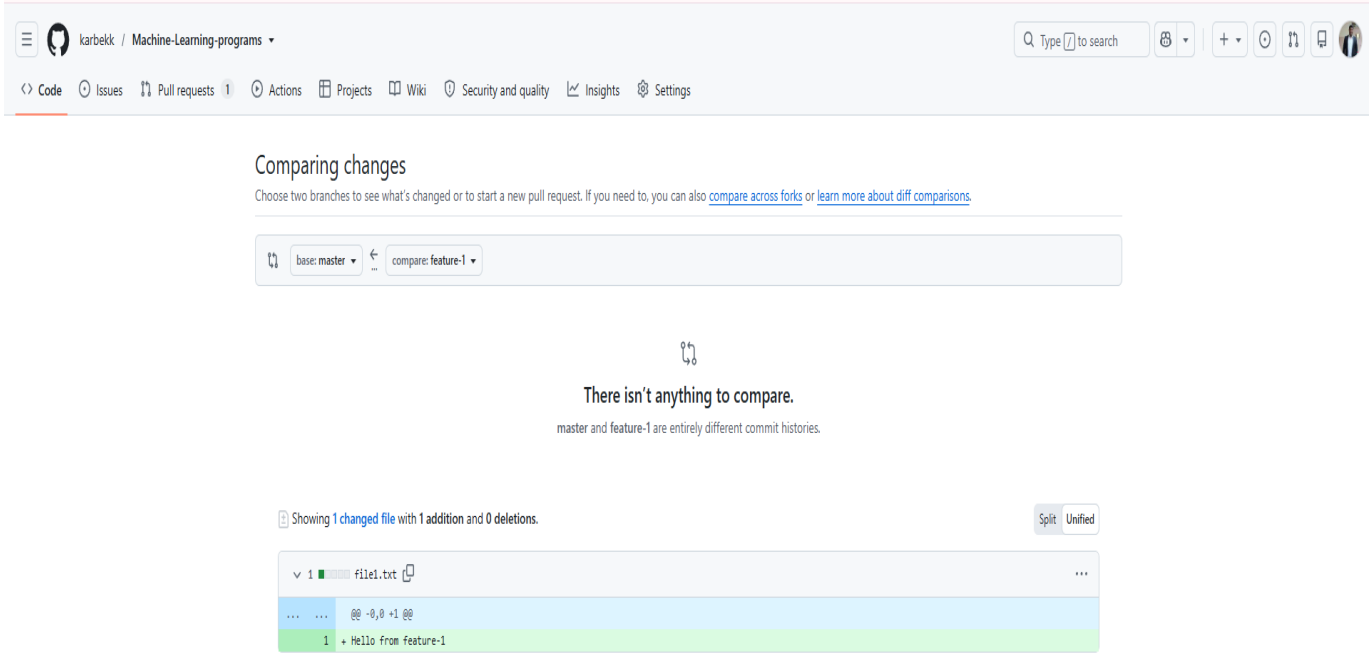
BMSIT@DESKTOP-9VOIESP MINGW64 ~ (feature-1)
$ git remote add origin https://github.com/karbekk/Machine-Learning-programs.g

BMSIT@DESKTOP-9VOIESP MINGW64 ~ (feature-1)
$ git remote -v
origin  https://github.com/karbekk/Machine-Learning-programs.git (fetch)
origin  https://github.com/karbekk/Machine-Learning-programs.git (push)
```

```
BMSIT@DESKTOP-9VOIESP MINGW64 ~ (feature-1)
$ git push -u origin feature-1
info: please complete authentication in your browser...
Enumerating objects: 18, done.
Counting objects: 100% (18/18), done.
Delta compression using up to 12 threads
Compressing objects: 100% (12/12), done.
Writing objects: 100% (18/18), 1.56 KiB | 800.00 KiB/s, done.
Total 18 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), done.
remote:
remote: Create a pull request for 'feature-1' on GitHub by visiting:
remote:      https://github.com/karbekk/Machine-Learning-programs/pull/new/feature-1
remote:
To https://github.com/karbekk/Machine-Learning-programs.git
 * [new branch]      feature-1 -> feature-1
branch 'feature-1' set up to track 'origin/feature-1'.
```

# Check in Github Repo for Compare and Pull request



# Step 4: Rebase your branch on the latest main

```
BMSIT@DESKTOP-9VOIESP MINGW64 ~ (feature-1)
$ git push -u origin feature-1
branch 'feature-1' set up to track 'origin/feature-1'.
Everything up-to-date

BMSIT@DESKTOP-9VOIESP MINGW64 ~ (feature-1)
$ git rebase origin/main
Current branch feature-1 is up to date.
Enumerating objects: 109, done.
Counting objects: 100% (109/109), done.
Delta compression using up to 12 threads
Compressing objects: 100% (85/85), done.
Writing objects: 100% (109/109), done.
Total 109 (delta 26), reused 0 (delta 0), pack-reused 0 (from 0)
Enumerating cruft objects: 408, done.
Traversing cruft objects: 408, done.
Counting objects: 100% (408/408), done.
Delta compression using up to 12 threads
Compressing objects: 100% (401/401), done.
Writing objects: 100% (408/408), done.
Total 408 (delta 84), reused 0 (delta 0), pack-reused 0 (from 0)

BMSIT@DESKTOP-9VOIESP MINGW64 ~ (feature-1)
$
```

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# Step 5: Merge with a custom commit message

```
BMSIT@DESKTOP-9VOIESP MINGW64 ~ (feature-1)
$ git push --force-with-lease origin feature-1
Everything up-to-date

BMSIT@DESKTOP-9VOIESP MINGW64 ~ (feature-1)
$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.

BMSIT@DESKTOP-9VOIESP MINGW64 ~ (main)
$ git pull origin main
From https://github.com/karbekk/Machine-Learning-programs
 * branch            main       -> FETCH_HEAD
Already up to date.

BMSIT@DESKTOP-9VOIESP MINGW64 ~ (main)
$ git merge --no-ff feature-1 -m "Merge feature-1: added file1"
Merge made by the 'ort' strategy.
 file1.txt | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 file1.txt

BMSIT@DESKTOP-9VOIESP MINGW64 ~ (main)
$ git push origin main
Enumerating objects: 1, done.
Counting objects: 100% (1/1), done.
Writing objects: 100% (1/1), 223 bytes | 223.00 KiB/s, done.
Total 1 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/karbekk/Machine-Learning-programs.git
   f34179a..66067c6  main -> main
```

Dr Kartik